

Proof P. 2.1: (h has multiple comps)

two issues with my understanding

H minimal el
 H contains the
 rightmost vertex of h
 $\Downarrow$

H is also maximal
 exists multiple
 minimal elements

if f is the outer-face
 it can be made
 inner face by adding
 an edge from x_n
 to the rightmost
 vertex of h

a Partial order $\prec$

- reflexive

$$H \prec H$$

- Antisymmetric

$$\text{if } H_1 \prec H_2$$

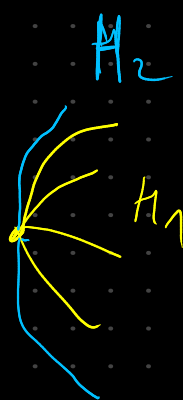
$$\text{not } H_2 \prec H_1$$

$$\Rightarrow H_1 = H_2$$

- transitive

$$\text{if } H_1 \prec H_2 \text{ and } H_2 \prec H_3 \\ \Rightarrow H_1 \prec H_3$$

a strict partial order:



these are not
 true in $\prec$

- irreflexivity $\neg (H_1 < H_1)$ true

- Asymmetry if $H_1 < H_2 \Rightarrow \neg (H_2 < H_1)$ true

- transitivity // //

minimal el H if $\forall H'' \in S \quad \underline{H < H''}$

$\neg (H < H)$

also H

$\uparrow$

$\forall x \in S \quad H < x$

} there is no el such that $x > H$

because $H > H \rightarrow \text{false}$

| there is no el such that $x < H$

since $H < H$
false

So if H is minimal (when its edges form H_0)
are consecutive in H_{k+1}

and if H has the right most vertex k .

all edge are even



then if $k \in H$ an edge
going through H linking
upper side to lower side of

H_2 with $H_2 \supset H$

(odd crossing) $\Rightarrow$ (contradiction)
So H is also maximal.

exist an
other minimal $\Leftarrow$

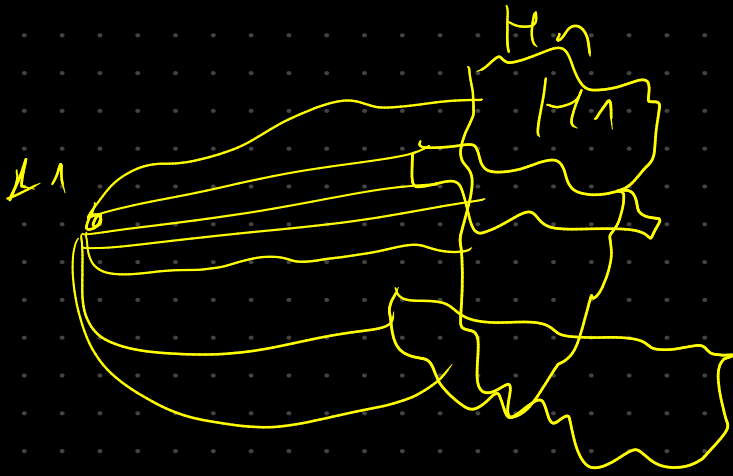
e)
(we have
more than
one component
in $H - \{k\}$)

$H' = H - \{x_1\}$ considering $h - V(H')$

the state before the induction



the state after adding x_1 and its edges on the right;



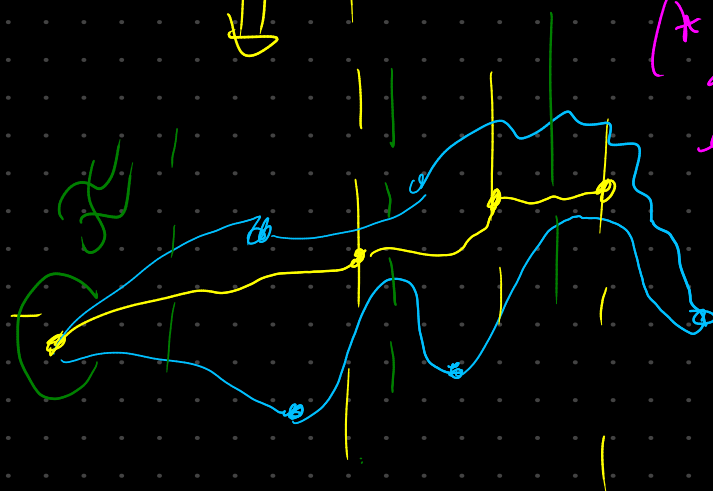
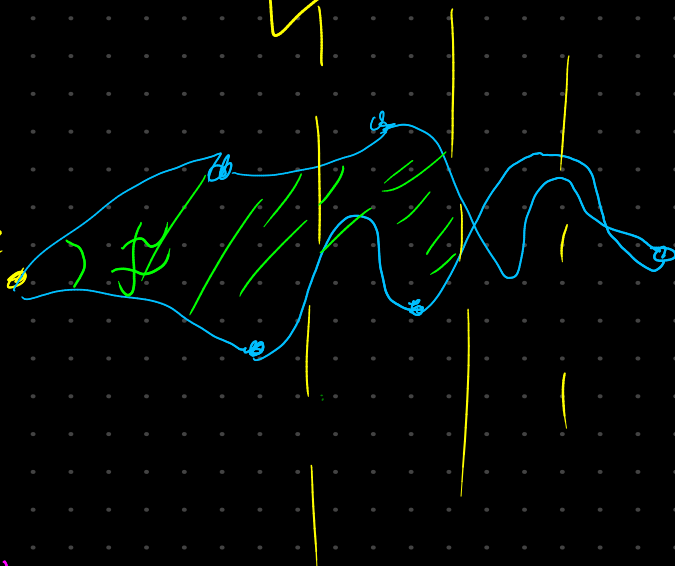
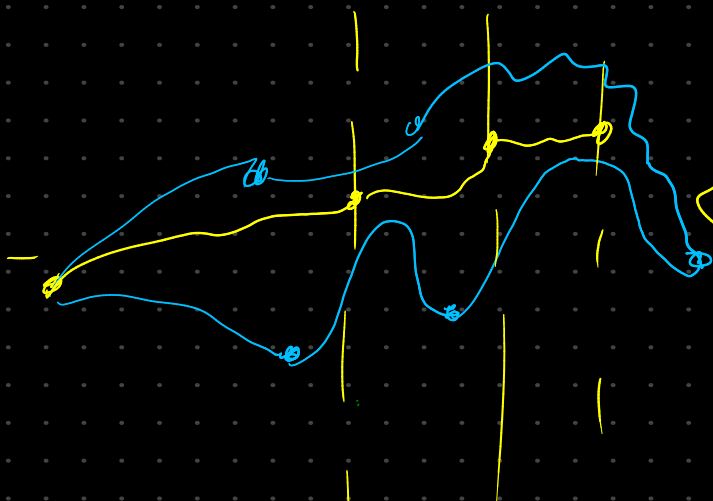
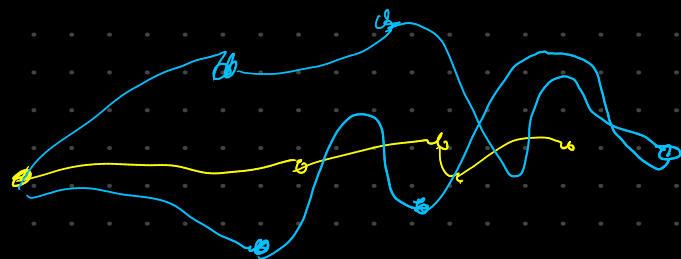
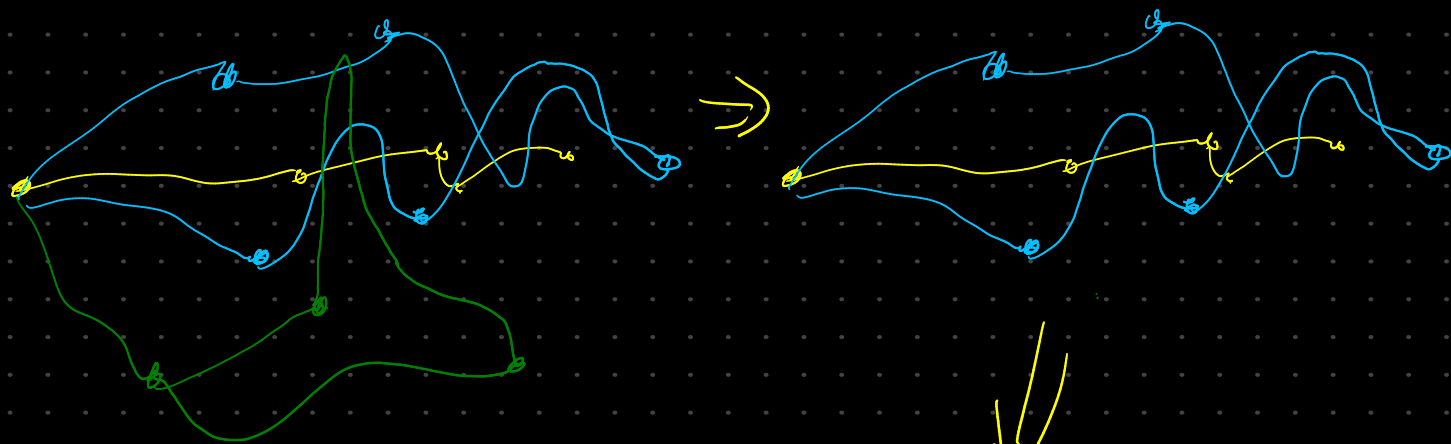
now we want to remove one component of $h - \{x_1\}$

(if we remove one by one)

we will end up with one.

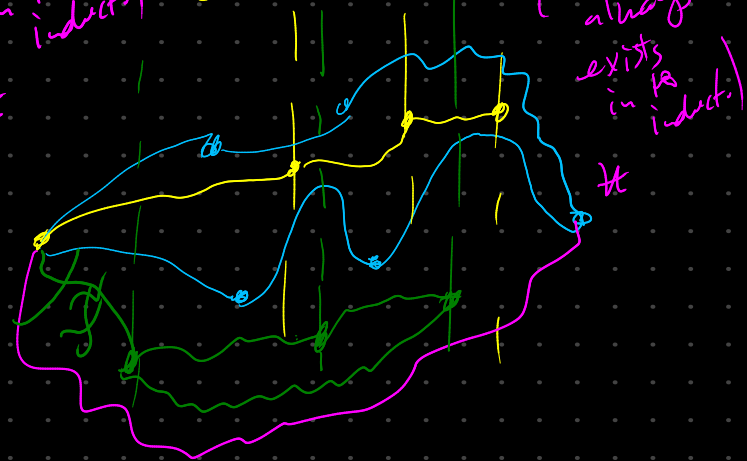
if I understand, we keep removing minimal
el of what's left of S

if the min has the right-most α we look for an other min and we keep going until we have just one.



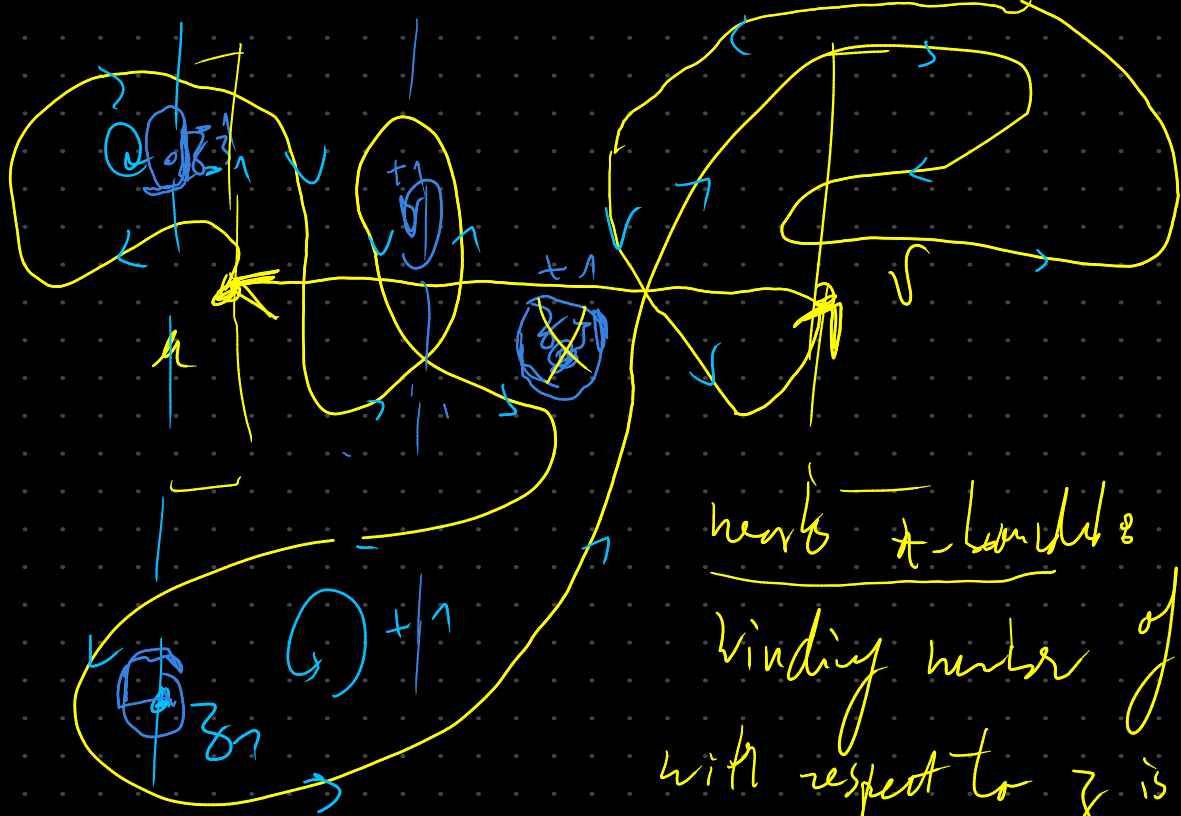
(α will always exist in the induct.)

α



(α will always exist in the induct.)

α



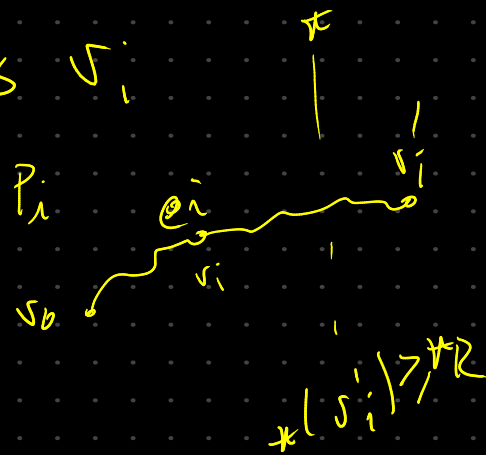
marks x -boundary
 winding number of C_e
 with respect to z is even.

$w(z_1) = 1$ (skipped for now)

Lemma 3.4

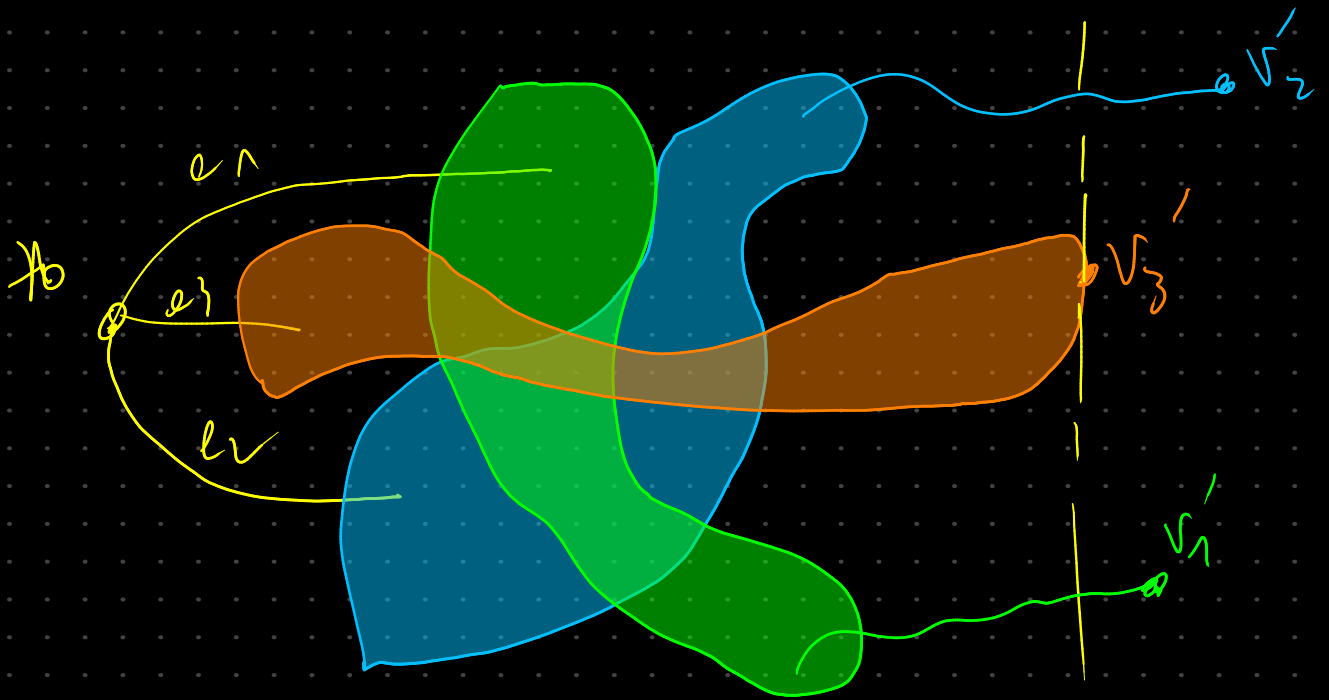
for an arbitrary $x_R > x(v_0)$, G' graph induced
 by h on vertices v with $x(v_0) < x(v) \leq x_R$.

G'_i be the comp. of G' that contains v_i



x_R

Suppose h_1, h_2, h_3 are pairwise disjoint and that for every i there is a path P_i (in h) from v_0 through e_i to some vertex v'_i satisfying $x(v'_i) \geq x_e$, so all v of P_i $x(v) \geq x(v_0)$. Then each h_i has no N_{ing} in h to the left of $x(v_0)$.



Making P_i to be minimal

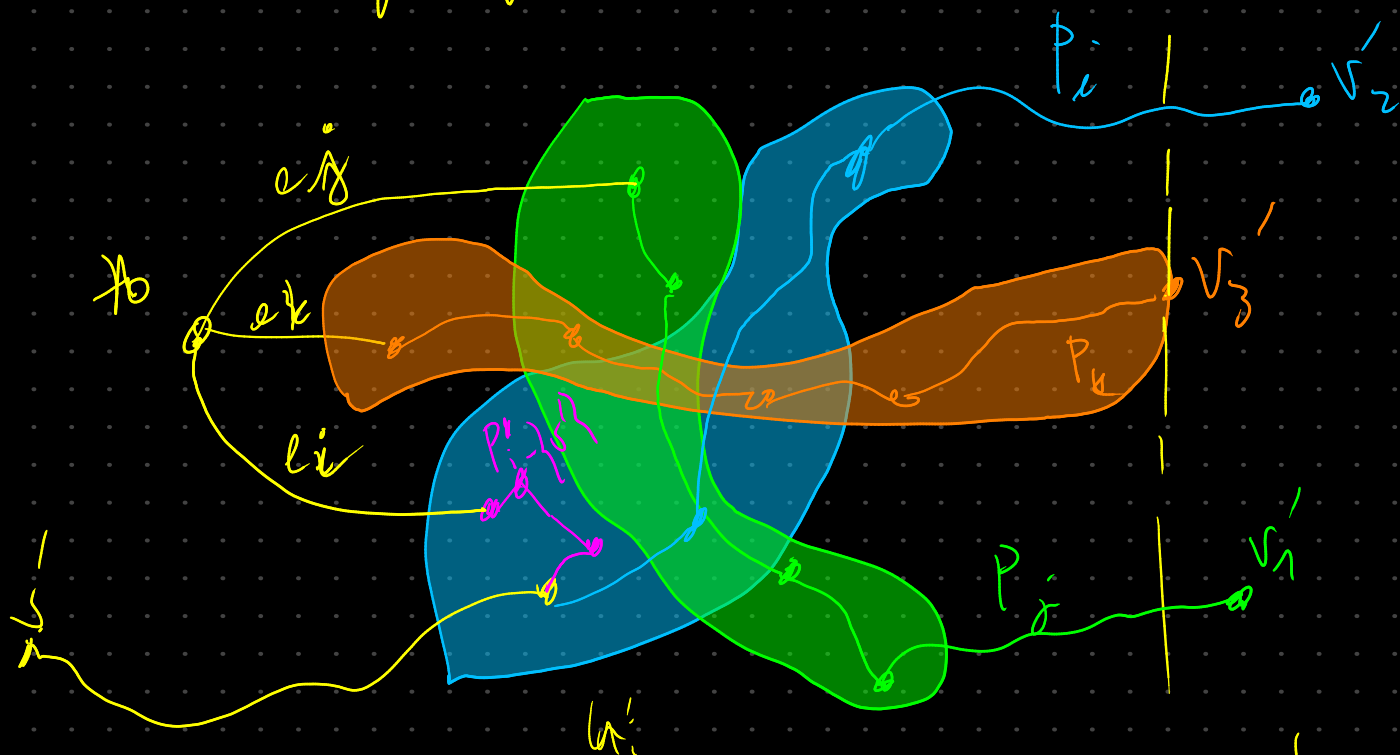
$$\forall v \in P_i - \{v'_i\} \quad x(v_0) \leq x(v) < x_e$$

$$\left\{ \begin{array}{l} \{u, v\} / x(u) \leq x_e < x(v) \\ q \subseteq H^2 \text{ and } q \neq P_i \end{array} \right\}$$

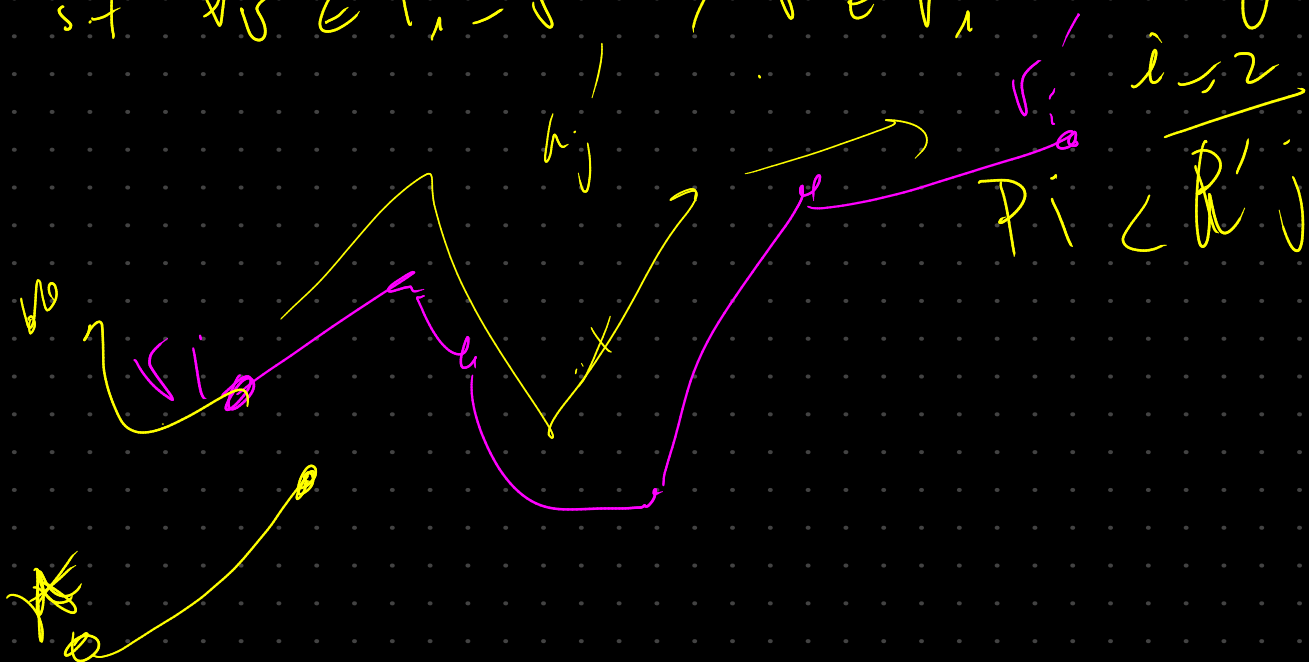
$$x(v'_i) \geq x_e$$

$\forall q \subseteq q$ is either $\rightarrow$ alone P_i
 $\searrow$ follow

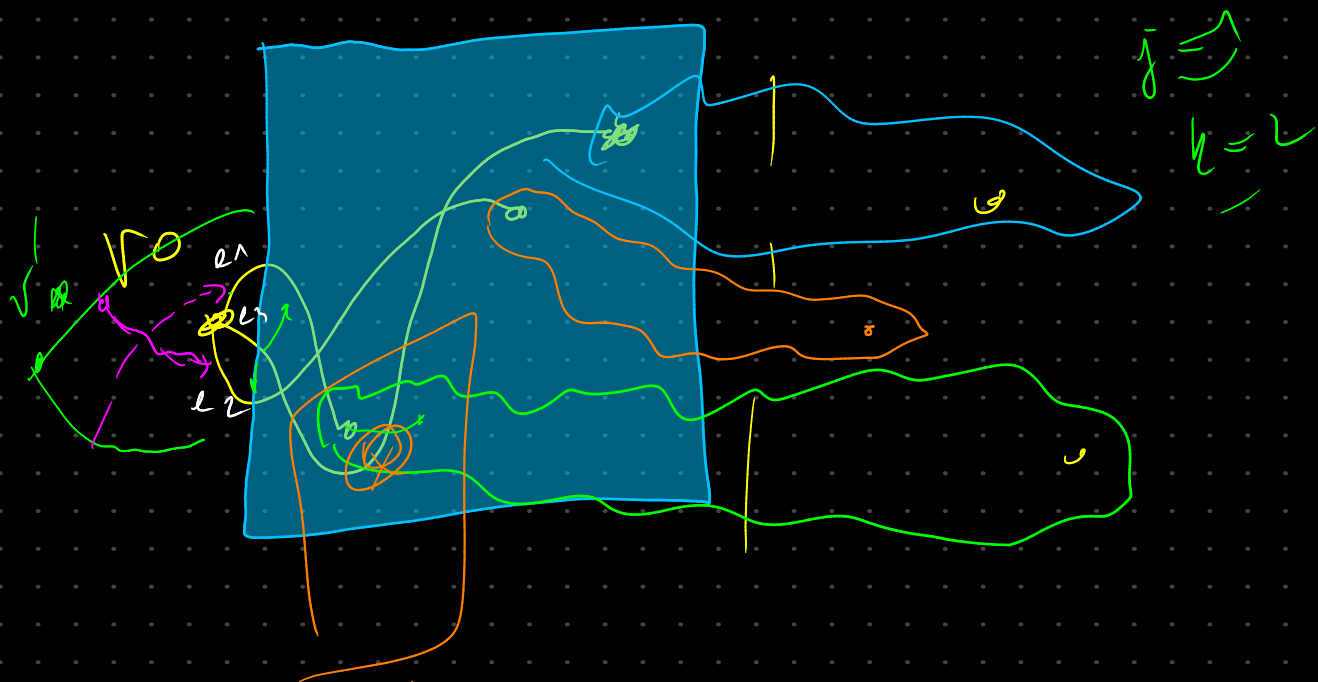
Suppose (contradiction) that u_i has a neighbor v' to the left of $x(v_0)$.



P_i' be a path from v_i to s
 s.t. $\forall v \in P_i' - s, v \in h_i'$ in my ex.



$\Rightarrow p'_i$ must cross x , both above and below x_0 (contradiction)



we can't
get away
with 10's
because of
 $\mathbb{R}$